Hyperspectral Intrinsic Image Decomposition With Enhanced Spatial Information

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***Abstract*— Hyperspectral intrinsic image decomposition (HyperIID) has been proven to be a very useful approach to reduce the spectral uncertainty in the remote sensing imaging process and improve the classification. In this article, a new HyperIID with enhanced spatial information, called ESI-IID, is proposed to overcome the deficiency of low spatial resolution in the existing HyperIID methods. With the aid of high-resolution (HR) panchromatic (PAN) image, the proposed method embeds the HR spatial information into the intrinsic decomposition model and enhances spatial details of the intrinsic component. The proposed ESI-IID introduces three constraints: 1) we make the constraint on spectral information to protect it from distortion during the spatial resolution enhancement process; 2) we add the constraint on spatial information to make sure that the details of edges will be well kept; and 3) based on the assumption that the reflectance component has a strong correlation in the local neighborhood, we add the self-constraint on reflectance component, in which the similarity matrix consists of two parts extracted from hyperspectral images and PAN image, respectively. Finally, we build a matrix energy function according to the aforementioned constraints and solve it by finding the minimum Frobenius norm iteratively. Both visual and quantitative experiments on simulated and real datasets demonstrate that the proposed method outperforms other alternative methods with high reliability.**

***Index Terms*— Classification, hyperspectral images (HSIs), image fusion, intrinsic image decomposition (IID), spatial information.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spectral information, which is useful for many tasks, such as environmental monitoring [1], target recognition [2], change detection [3], and classification [4]. Particularly, HSI classification has been a vibrant research topic in recent years. It is a challenging task due to the existence of redundant features, mixed pixels, and high-dimensional spectral data in HSIs [5]. Researches indicate that feature extraction is an effective way to improve the classification accuracy of HSIs, and researchers have developed a variety of feature extraction methods for HSIs' classification with fully considering the spatial and spectral information of HSIs [6]–[8]. In addition, HSI unmixing and subpixel mapping methods have also been extensively studied for this purpose [9]–[12].

Recently, intrinsic image decomposition (IID) has attracted much attention in the field of computer vision that aims at separating the reflectance and shading components from a single image [13]. As the reflectance component corresponds to the material-dependent properties of different objects, IID can benefit many classic computer vision problems, such as semantic segmentation and object detection [14], [15]. IID is still a challenging problem although many methods have been proposed to solve it, such as object-based methods [16], scene-based methods [17], and image sequences-based methods [18], [19]. Kang *et al.* [20] first introduced the IID technology into HSIs as an effective feature extraction method by three steps, called the GroupIID method, and their experimental results showed that hyperspectral intrinsic image decomposition (HyperIID) is a very powerful approach to improve the classification.

Pansharpening is of great significance in remote sensing image processing that aims at fusing a panchromatic (PAN) image with a multispectral image (MSI) to generate an image with both high spatial resolution and high spectral resolution [21]. Many algorithms have been proposed to solve this problem for the last decade, and they can be divided into four classes [21]: component substitution (CS), multiresolution analysis (MRA), Bayesian methods [22], [23], and variational methods [24], [25]. The CS methods solve the pansharpening problem by separating the spatial and spectral information in another space and then substituting the PAN image for components that contain spatial structure, and it includes many popular approaches, such as the principal component analysis (PCA) [26], Brovey [27], [28], and Gram–Schmidt (GS) methods and its improved version called GS Adaptive (GSA) [29], [30]. MRA-based methods first extract the spatial details from PAN image via multiresolution decomposition and then inject the details into the resampled MSI. The representative MRA-based methods include smoothing filter-based intensity modulation (SFIM) and Laplacian pyramid (LP) [31], [32]. The Bayesian methods are implemented by finding a suitable statistical model for the pansharpened result and the input images. As for the variational methods, they achieve the pansharpening task by solving an optimization problem with predefined prior knowledge. Based on the linear combination assumption, Ballester *et al.* [24] first proposed a variational model named P+XS, which is an innovation in remote sensing image fusion. Following the P+XS method, an alternate

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variational wavelet pansharpening (AVWP) method was presented by adding a spectral ratio constraint and assuming that the wavelet fused image is closed to the pansharpened MSI [33]. More recently, Liu *et al.* [34] proposed an intelligent spatial-Hessian-feature-guided variational pansharpening method, in which the intracorrelations across different channels can be fully used and more elaborate geometric structure can be preserved; experimental results demonstrated that it outperforms various pansharpening methods. To better preserve image details and texture, an efficient method was proposed by integrating spatial fractional-order gradient feature and nonlocal patch-based low-rank prior [35], which can achieve promising results.

With the increasing applications of hyperspectral imaging systems, some problems of HSIs are gradually exposed. HSIs have the characteristics of high spectral resolution, which benefits varieties of remote sensing applications [36]–[39]. However, HSIs are relatively restricted in terms of spatial resolution due to the physical and technical bottlenecks of imaging spectrometers [40]. To solve this problem, a number of pansharpening methods have been extended to HSIs [41]–[43]. Methods designed for hyperspectral pansharpening can be broadly divided into Bayesian methods and matrix factorization methods. The Bayesian class contains algorithms such as naive Gaussian prior method [44], sparsity promoting prior method [45], and Hysure [46]. Coupled nonnegative matrix factorization (CNMF) is the representative method of the matrix factorization class that realizes the fusion task by alternately unmixing both sources of data to obtain the endmember spectra and the high spatial resolution abundance maps [47]. Recently, deep learning has developed rapidly, and it has been successfully applied in many fields. Palsson *et al.* [48] proposed a method that uses a 3-D convolutional neural network to fuse HSI and MSI. Qu *et al.* [49] well achieved the super-resolution issue by using unsupervised encoder–decoder architecture (USDN), which consists of two sparse Dirichlet-Nets to, respectively, extract the spectral and spatial information from HSI and MSI, and outperform the state of the arts.

Besides, single image super-resolution methods are also introduced into this field to address the aforementioned problem. Yan *et al.* [50] proposed an edge interpolation-based method. There is also a group of methods based on sparse coding and its improvements [51], [52], in which the patches are the focus of the optimization. As a result of the good performance in many computer vision tasks, deep learning methods have also been widely used in this area [53]. However, due to the high-dimensional patterns and requirements of many training images, both the deep learning and dictionary-based methods for single HSI need to be further developed. In addition, hybrid methods combined with single image super-resolution methods and fusion methods are also proposed to enhance HSI [54]. For instance, Kwan *et al.* [55] developed an efficient method by integrating two recent image processing techniques, namely, the plug-and-play algorithm and hybrid color mapping, which performs deblurring and fusion steps, respectively. Comprehensive results have shown its promising ability.

In this article, a novel HyperIID with enhanced spatial information, called ESI-IID, is proposed to break the limitation of low spatial resolution to the existing HyperIID methods. The proposed ESI-IID method is based on three reasonable constraints, i.e., constraints on spatial information, spectral information, and reflectance component, respectively. The first constraint is considered from the perspective of spectral attributes, which aims at protecting spectral information from distortion; the second is considered from the perspective of spatial attributes, which can keep the details of edges very well. Furthermore, the third one is to maintain the relationship between the reflectance of each pixel, which is obtained from HSIs and PAN data. Finally, we find the minimum Frobenius norm of the proposed matrix energy function through iteration.

The main contributions of this work are given as follows.

1) The high resolution of spatial information provided by the PAN image is exploited to improve the spatial resolution of the target reflectance image. To the best of our knowledge, it is the first work to improve the spatial resolution during the IID process of HSIs.
2) Instead of being estimated from an artificially generated fused image, the similarity matrix is estimated from the real relationship within the pixels of HSIs and that within PAN data, which provides more reliable results.

The rest of this article is organized as follows. In Section II, we review the related works used in HyperIID methods. Section III describes the proposed algorithm based on three constraints in detail. The experimental results and analysis are shown in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Works

A. Basic Model of HyperIID

In the field of computer vision, the intrinsic images decomposition problem for red-green-blue (RGB) images can be described as the following equation:

$$\mathbf{I} = \mathbf{R} \circ \mathbf{S} \tag{1}$$

where $\mathbf{I}$ represents the original image, $\mathbf{R}$ represents the reflectance component, $\mathbf{S}$ represents the shading component, and $\circ$ represents the elementwise multiplication operator. Consider each pixel i in the image $\mathbf{I}$; here, we have $\mathbf{I}_i = (\mathbf{I}_{ir}, \mathbf{I}_{ig}, \mathbf{I}_{ib})$, $\mathbf{R}_i = (\mathbf{R}_{ir}, \mathbf{R}_{ig}, \mathbf{R}_{ib})$, and r, g, and b denote the RGB channels of a color image, respectively. Meanwhile, the shading component here is not the function of wavelength.

Unlike the above assumptions for RGB images, the illumination in HSIs is the result of the interaction between distant sunlight and the atmosphere. Therefore, the shading component of HSIs affects each wavelength differently. Then, the IID model for HSIs is given as follows:

$$\mathbf{I}(\lambda) = \mathbf{R}(\lambda) \circ \mathbf{S}(\lambda) \tag{2}$$

where λ is the wavelength.

Generally speaking, there are two main representative methods in the current HyperIID methods: the GroupIID method and the HIID method.

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B. Group IID

As for the GroupIID method, they first reduce the spectral dimension of the HSIs and then divide the reduced image $\bar{\mathbf{I}}$ into several subsets of adjacent bands as follows:

$$\hat{\mathbf{I}}^k = \begin{cases} \left(\bar{\mathbf{I}}_{(k-1)Z+1}, \ldots, \bar{\mathbf{I}}_{(k-1)Z+Z}\right), & k = 1, 2, \ldots, \left\lfloor \dfrac{M}{Z} \right\rfloor \\ \left(\bar{\mathbf{I}}_{M-Z+1}, \ldots, \bar{\mathbf{I}}_{M}\right), & k = \lceil \dfrac{M}{Z} \rceil \neq \left\lfloor \dfrac{M}{Z} \right\rfloor \end{cases} \tag{3}$$

where $\hat{\mathbf{I}}^k$ refers to the kth group, $\lfloor M/Z \rfloor$ is the largest integer not greater than M/Z, $\lceil M/Z \rceil$ is the smallest integer not less than M/Z, and Z represents the number of bands of each subgroup.

Finally, they use an optimized IID technology to estimate the reflectance component of each subset. Formally, the GroupIID problem formulation is

$$E\left(\mathbf{R}^k, \mathbf{S}^k\right) = \sum_{i \in \hat{\mathbf{I}}^k} \left(\mathbf{R}_i^k - \sum_{j \in \omega_i^k} \alpha_{ij}^k \mathbf{R}_j^k \right)^2 + \sum_{i \in \hat{\mathbf{I}}^k} \left(\frac{\hat{\mathbf{I}}^k}{\mathbf{S}_i^k} - \mathbf{R}_i^k \right)^2 \tag{4}$$

where $\mathbf{R}^k$ and $\mathbf{S}^k$ refer to the kth grouped reflectance and shading component of the dimension reduced HSI, respectively, the first term is the cost function for reflectance, and the second represents the data fidelity term.

Experimental results show that the GropIID technology is an effective way to extract features from HSIs. However, this algorithm reduces the spectral resolution of HSI for more effective computer processing, which may lead to difficulties in detecting and distinguishing various objects.

C. HIID

Jin *et al.* [56] proposed a HyperIID method called HIID with two additional priors, namely, the reflectance prior and the shading prior, which shows excellent performance in the HSIs' classification task. Similarly, the problem formulation consists of two cost functions

$$E\left(\boldsymbol{\rho}, \tilde{\boldsymbol{\beta}}, \mathbf{L}\right) = \left\| \boldsymbol{\rho} + \mathbf{L}\mathbf{1}_n^T + \mathbf{1}_d \tilde{\boldsymbol{\beta}} - \mathbf{O} \right\|_F^2 + \frac{1}{\sigma} \left\| \boldsymbol{\rho} - \boldsymbol{\rho}\mathbf{W}^T \right\|_F^2 \tag{5}$$

where $\boldsymbol{\rho}$ is a log-reflectance image, $\tilde{\boldsymbol{\beta}}$ is a log-orientation image, $\mathbf{L}$ is a log-illumination image, $\mathbf{1}_n$ and $\mathbf{1}_d$ denote the $n \times 1$ and $d \times 1$ vectors with all 1 s, respectively, and $\mathbf{O}$ is the log-HSI.

In the first data fidelity term, the shading component in the observation model has been further decomposed into such two parts as environment illumination and surface orientation changes in the HIID method by using a shading prior to reduce the number of unknowns. Besides, in the second cost function, a reflectance prior that extracts embedding structure of same material based on L1-graph coding is used. By contrast, the HIID method has more advantages in keeping spectral continuity and improving the accuracy of reflectance recovery than the GroupIID method since there is no process of reducing spectral dimension and the prior on reflectance is more reasonable.

Though the experimental results of the aforementioned HyperIID methods show that they perform well on reflectance recovery tasks, they cannot solve the problem that HSIs are practically restricted in terms of spatial resolution and spectral aliasing. To address these problems, we will introduce a high-resolution (HR) PAN image as the auxiliary data and generate a reflectance component with enhanced spatial resolution compared with input low-resolution HSI (LR-HSI) by constructing three reasonable constraints. We call our proposed technique "ESI-IID."

III. Proposed ESI-IID

In this article, our aim is to decompose HR reflectance $\mathbf{R}_h$ and shading $\mathbf{S}_h$ component from original low resolution HSI $\mathbf{I}_l$ with the help of PAN image $\mathbf{P}$. For convenience, we transform (5) into log domain and introduce new variables $\tilde{\mathbf{I}} = \log(\mathbf{I}_l)$, $\tilde{\mathbf{R}} = \log(\mathbf{R}_h)$, $\tilde{\mathbf{S}} = \log(\mathbf{S}_h)$, and $\tilde{\mathbf{P}} = \log(\mathbf{P})$. Moreover, we define the total number of pixels in $\tilde{\mathbf{P}}$ and the number of channels in $\tilde{\mathbf{I}}$ as n and d, respectively. And the schematic diagram of the proposed method is shown in Fig. 1.

A. Constraint on Spectral Information

Considering spatial degradation, low-resolution pixels are generated by HR ones through low-pass filtering and downsampling. The spectral information at high spatial resolution should remain the same as that in LR-HSI after the spatial degradation. Based on this assumption, the relationship of reflectance and shading component in high spatial resolution and original LR-HSI can be formulated as

$$U\tilde{\mathbf{I}} = L\left(\tilde{\mathbf{R}} + \tilde{\mathbf{S}}\right) + N_1 \tag{6}$$

where $N_1 \in \mathbb{R}^{n \times d}$ represents additive noise, U denotes the bicubic interpolation upsampling operator, and L is a cyclic convolution operator acting on the bands that model the point spread function (PSF) of the hyperspectral sensor, which is viewed as a blurring-decimation operator and defined as a Gaussian kernel in previous articles [22], [57]–[59]. In this work, we define L as a Gaussian kernel of size 3×3.

Therefore, the spectrum-consistent term can be written as

$$E_1\left(\tilde{\mathbf{R}}, \tilde{\mathbf{S}}\right) = \left\| U\tilde{\mathbf{I}} - L\left(\tilde{\mathbf{R}} + \tilde{\mathbf{S}}\right) \right\|_F^2 \tag{7}$$

where $\| \cdot \|_F$ denotes the Frobenius norm.

B. Constraint on Spatial Information

It is well known that the spatial information of high spatial resolution images is contained in the PAN image. Thus, we assume that the spatial structure information of HR-HSI is close to that of PAN image, and the amount of spatial detail required for different bands is proportional to the information ratio of the corresponding bands, which helps to reduce the spectral distortion. Generally, the spatial information of remote sensing images can be represented by the measure of gradient [60]. Therefore, based on this assumption, we can get the following equation:

$$\mathbf{T} \circ \nabla_2 \tilde{\mathbf{P}}_d = \nabla_2\left(\tilde{\mathbf{R}} + \tilde{\mathbf{S}}\right) + N_2 \tag{8}$$

$$\mathbf{T}(i) = \frac{\widehat{\mathbf{I}}(i)}{c_i \sum_{i=1}^{d} \widehat{\mathbf{I}}(i)} \tag{9}$$

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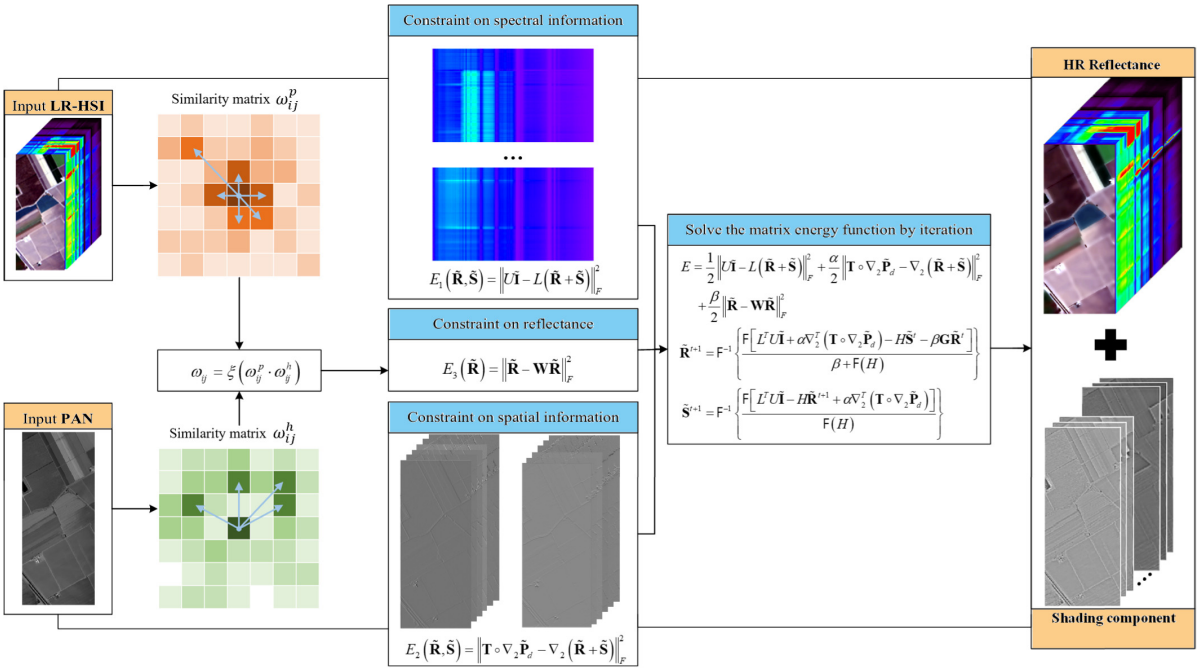


Fig. 1. Flowchart of the proposed method.

where $\tilde{\mathbf{P}}_d \in \mathbb{R}^{n\times d}$ represents $\tilde{\mathbf{P}}$ stacked d times in the spectral dimension, $\mathbf{T}$ denotes the ratio matrix of the spectral contribution of LR-HSI, c_i represents the coefficients related to the modulation transfer function (MTF) of the sensor [61], the division is a elementwise operator, $\widehat{\mathbf{I}}$ is the upsampled LR-HSI image by bicubic interpolation upsampling operator, $N_2 \in \mathbb{R}^{n\times d}$ represents additive noise, and the gradient operator $\nabla_2 = [\nabla_{2x}, \nabla_{2y}]$ can be calculated by the forward difference between adjacent pixels.

Therefore, the structure-consistent term can be written as

$$E_2(\tilde{\mathbf{R}},\tilde{\mathbf{S}}) = \left\|\mathbf{T}\circ\nabla_2\tilde{\mathbf{P}}_d - \nabla_2(\tilde{\mathbf{R}}+\tilde{\mathbf{S}})\right\|_F^2. \tag{10}$$

C. Constraint on Reflectance

Based on the assumption that the reflectance component has a strong correlation in the local neighborhood, we assume that the reflectance of one pixel is equal to the linear combination of the reflectance of its neighboring pixels

$$\tilde{\mathbf{R}}_i = \sum_{j=1}^{n} \omega_{ij}\tilde{\mathbf{R}}_j \tag{11}$$

where ω_{ij} measures the similarity between the pixel i and pixel j, so ω_{ij} ranges from 0 to 1. Inspired by [62], we assume that ω_{ij} consists of two parts, which are obtained from LR-HSI and PAN, respectively.

According to the Retinex theory [63]–[66], the shading component varies smoothly, whereas the reflectance component is piecewise constant where discontinuities correspond to the sharp edges in the image. Since the PAN image has a higher spatial resolution, the similarity matrix characterizing the discontinuities of reflectance is extracted from it, which can be expressed as follows:

$$\omega_{ij}^{p} = \begin{cases} e^{-(\nabla\mathbf{P}_{ij})^2/\sigma_{ip}^2}, & |\nabla\mathbf{P}_{ij}| > \tau \\ 1, & |\nabla\mathbf{P}_{ij}| \leqslant \tau \end{cases} \tag{12}$$

where τ is the threshold, $\nabla\mathbf{P}_{ij} = \mathbf{P}_i - \mathbf{P}_j$, and σ_{ip}^2 is the variance of pixel values related to PAN image. If the ith and jth pixels are not similar, we will keep the structural relationship between them; otherwise, the similarity will be measured by spectrum.

Based on the assumption that the pixels with similar spectral intensity values tend to share similar reflectance, the second matrix is extracted from LR-HSI based on spectral similarity, which can be written as

$$\omega_{ij}^{h} = \exp\left(-A\left(\widehat{\mathbf{I}}_i,\widehat{\mathbf{I}}_j\right)^2/\sigma_{ih}^2\right) \tag{13}$$

$$A\left(\widehat{\mathbf{I}}_i,\widehat{\mathbf{I}}_j\right) = \arccos\left[\left(\widehat{\mathbf{I}}_i^{\,T}\widehat{\mathbf{I}}_j\right)/\left(\sqrt{\widehat{\mathbf{I}}_i^{\,T}\widehat{\mathbf{I}}_i}\sqrt{\widehat{\mathbf{I}}_j^{\,T}\widehat{\mathbf{I}}_j}\right)\right] \tag{14}$$

where $\widehat{\mathbf{I}}_i, \widehat{\mathbf{I}}_j$ represent the intensity of the ith and jth pixels in upsampled LR-HSI image, respectively, and σ_{ih}^2 is the variance of angle values related to the LR-HSI image.

The final ω_{ij} is defined as

$$\omega_{ij} = \xi\left(\omega_{ij}^{p}\cdot\omega_{ij}^{h}\right) \tag{15}$$

where ξ is the normalization coefficient.

According to the above equations, the cost function based on (11) is given as follows:

$$E_3(\tilde{\mathbf{R}}) = \left\|\tilde{\mathbf{R}} - \mathbf{W}\tilde{\mathbf{R}}\right\|_F^2 \tag{16}$$

where $\mathbf{W} \in \mathbb{R}^{n\times n}$ is a normalized constraint matrix and $\mathbf{W}_{ij} = \omega_{ij}$.

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D. Solution of the Problem

To combine the above three cost functions together, we get a new optimization function

$$E = \frac{1}{2}\left\|U\tilde{\mathbf{I}} - L\left(\tilde{\mathbf{R}} + \tilde{\mathbf{S}}\right)\right\|_F^2 + \frac{\beta}{2}\left\|\tilde{\mathbf{R}} - \mathbf{W}\tilde{\mathbf{R}}\right\|_F^2 + \frac{\alpha}{2}\left\|\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d - \nabla_2\left(\tilde{\mathbf{R}} + \tilde{\mathbf{S}}\right)\right\|_F^2 \tag{17}$$

$$\min_{\tilde{\mathbf{R}},\tilde{\mathbf{S}}} E\left(\tilde{\mathbf{R}}, \tilde{\mathbf{S}}\right). \tag{18}$$

Given E, the first partial derivatives of E are given as follows:

$$\frac{\partial E}{\partial \tilde{\mathbf{R}}} = \left(L^T L + \alpha\nabla_2^T\nabla_2 + \beta\mathbf{I}_n + \beta\mathbf{G}\right)\tilde{\mathbf{R}} + \left(L^T L + \alpha\nabla_2^T\nabla_2\right)\tilde{\mathbf{S}} - L^T U\tilde{\mathbf{I}} - \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) \tag{19}$$

$$\frac{\partial E}{\partial \tilde{\mathbf{S}}} = \left(L^T L + \alpha\nabla_2^T\nabla_2\right)\left(\tilde{\mathbf{S}} + \tilde{\mathbf{R}}\right) - L^T U\tilde{\mathbf{I}} - \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) \tag{20}$$

$$\mathbf{G} = \mathbf{W}^T\mathbf{W} - \mathbf{W}^T - \mathbf{W} \tag{21}$$

where $\mathbf{I}_n$ denotes an $n \times n$ unit matrix.

By simply letting

$$\begin{cases} \partial E/\partial\tilde{\mathbf{R}} = \mathbf{0}_{n\times d} \\ \partial E/\partial\tilde{\mathbf{S}} = \mathbf{0}_{n\times d} \end{cases} \tag{22}$$

we get

$$\left(L^T L + \alpha\nabla_2^T\nabla_2 + \beta\mathbf{I}_n\right)\tilde{\mathbf{R}}^{t+1} = L^T U\tilde{\mathbf{I}} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) - \left(L^T L + \alpha\nabla_2^T\nabla_2\right)\tilde{\mathbf{S}}^t - \beta\mathbf{G}\tilde{\mathbf{R}}^t \tag{23}$$

$$\left(L^T L + \alpha\nabla_2^T\nabla_2\right)\tilde{\mathbf{S}}^{t+1} = L^T U\tilde{\mathbf{I}} - \left(L^T L + \alpha\nabla_2^T\nabla_2\right)\tilde{\mathbf{R}}^{t+1} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) \tag{24}$$

where t denotes the current number of iterations. Due to the existence of gradient operators in (23) and (24), we operate in the frequency domain to find the minimum Frobenius norm solution more efficiently by using the fast Fourier transform (FFT), which is achieved simply by carrying out the Fourier transform simultaneously on both sides of (23) and (24). Then, we get the iterative solution

$$\tilde{\mathbf{R}}^{t+1} = F^{-1}\left\{\frac{F\left[L^T U\tilde{\mathbf{I}} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) - H\tilde{\mathbf{S}}^t - \beta\mathbf{G}\tilde{\mathbf{R}}^t\right]}{\beta + F(H)}\right\} \tag{25}$$

$$\tilde{\mathbf{S}}^{t+1} = F^{-1}\left\{\frac{F\left[L^T U\tilde{\mathbf{I}} - H\tilde{\mathbf{R}}^{t+1} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right)\right]}{F(H)}\right\} \tag{26}$$

$$H = L^T L + \alpha\nabla_2^T\nabla_2 \tag{27}$$

$$F(H) = \overline{F(L)} \circ F(L) + \alpha F\left(\nabla_2^T\nabla_2\right) \tag{28}$$

where $F(\cdot)$ and $F^{-1}(\cdot)$ denote the FFT operator and its inverse, respectively. Using (25)–(28), the optimization function E can converge to $E(\tilde{\mathbf{R}}^*, \tilde{\mathbf{S}}^*)$, and the proposed algorithm is shown in Algorithm 1.

Algorithm 1 Overall Procedure for the Proposed Method

Input: LR-HSI image $\mathbf{I}$, and PAN image $\mathbf{P}$.

Compute the constraint matrix $\mathbf{W}$.

$$\mathbf{W}_{ij} = \xi\left(\omega_{ij}^p \cdot \omega_{ij}^h\right)$$

$$\omega_{ij}^p = \begin{cases} e^{-\left(\nabla\mathbf{P}_{ij}\right)^2/\sigma_{ip}^2} & \left|\nabla\mathbf{P}_{ij}\right| > \tau \\ 1 & \left|\nabla\mathbf{P}_{ij}\right| \leqslant \tau \end{cases}$$

$$\omega_{ij}^h = \exp\left(-A\left(\hat{\mathbf{I}}_i, \hat{\mathbf{I}}_j\right)^2/\sigma_{ih}^2\right)$$

Initialize: the reflectance matrix $\tilde{\mathbf{R}}$, and the shading matrix $\tilde{\mathbf{S}}$ as:

$$\tilde{\mathbf{R}}^0 = \log\left(\hat{\mathbf{I}}\right),\ \tilde{\mathbf{S}}^0 = \mathbf{0}^{n\times d}$$

repeat

  Find the optimal solution $\tilde{\mathbf{R}}$ by

$$\tilde{\mathbf{R}}^{t+1} = F^{-1}\left\{\frac{F\left[L^T U\tilde{\mathbf{I}} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right) - H\tilde{\mathbf{S}}^t - \beta\mathbf{G}\tilde{\mathbf{R}}^t\right]}{\beta + F(H)}\right\}$$

  Find the optimal diagonal matrix $\tilde{\mathbf{S}}$ by

$$\tilde{\mathbf{S}}^{t+1} = F^{-1}\left\{\frac{F\left[L^T U\tilde{\mathbf{I}} - H\tilde{\mathbf{R}}^{t+1} + \alpha\nabla_2^T\left(\mathbf{T} \circ \nabla_2\tilde{\mathbf{P}}_d\right)\right]}{F(H)}\right\}$$

  update iteration: $t = t + 1$.

until $\frac{\left\|\tilde{\mathbf{R}}^{t+1} - \tilde{\mathbf{R}}^t\right\|}{\left\|\tilde{\mathbf{R}}^t\right\|} < \varepsilon$;

Output: high-resolution reflectance $\exp\left(\tilde{\mathbf{R}}^*\right)$.

IV. Experimental Results and Analysis

In this section, some experiment results are reported to illustrate the effectiveness of the proposed method.

A. Data Description and Experimental Setting

The experiments are validated on three datasets including two simulated datasets and one real dataset: 1) the Salinas image; 2) the Botswana image; and 3) the Yellow River image. The first two datasets are treated as the HR-HSIs, and the PAN images and corresponding LR-HSIs at reduced resolution are generated in the same way, as illustrated in [67]. Specifically, the degradation is obtained by applying to the available HR-HSIs a low-pass filter (LPF) and a decimation operator, and the sampling factor is equal to the scale ratio r.

1) Salinas: This HSI was acquired by the AVIRIS sensor over the Salinas Valley in California, USA. The AVIRIS sensor operates in the wavelength range of 0.4–2.5 mm, and its spatial resolution has reached 3.7 m. The size of the image is 512 × 216 pixels, and the image originally has 224 bands; we use the image of 204 bands left after excluding 20 bands that cannot be reflected by water. Fig. 2(a) shows false-color images of this dataset.

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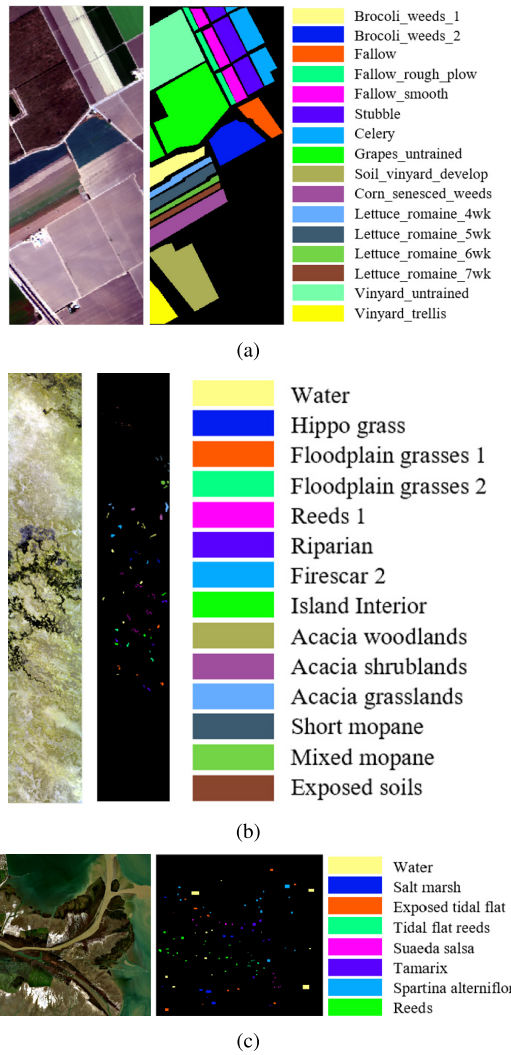


Fig. 2. Ground reference maps for three datasets. (a) Salinas. (b) Botswana. (c) Yellow River.

2) Botswana: This HSI with 1476×256 pixels was acquired by the Hyperion sensor over the Okavango Delta, Botswana. The Hyperion acquires 224 bands at 30-m spatial resolution in 10-nm bands from 400 to 2500 nm, and we use the image of 145 bands left after excluding 79 bands [10–55, 82–97, 102–119, 134–164, and 187–220] that uncalibrated and noisy bands. Fig. 2(b) shows false-color images of this dataset.

3) Yellow River: The HSI was captured by gaofen-5 (GF-5) above the Yellow River Wetland Park on November 1, 2018. The PAN covering the same area is generated from the MSI obtained by GF-1 wide field of view (WFV), in which the acquisition time is October 31, 2018. In only one day interval between the two acquisitions, we assume that there is no change in land cover during this period. The GF-5 acquires 330 bands at 30-m spatial resolution from 390 to 2500 nm, and we use the image of 118 bands left after removing 212 noisy bands [1–5, 87–89, 127–330]. The GF-1 WFV acquires four bands at the 16-m spatial resolution, and the PAN image was generated from it, as illustrated in [67]. Fig. 2(c) shows false-color images of this dataset.

Table I provides the detailed information of samples.

The parameters of our method are set as follows: the scale ratio r for Salinas and Botswana datasets is set as 4, the threshold factor τ is set as 0.06, the parameter ε in stop criterion is set as 0.001, and the weight coefficients are $\alpha \in [5, 15]$ and $\beta \in [1, 10]$.

In order to quantitatively verify the effectiveness of the proposed method, we use the standard K-nearest neighbor (KNN) classifier to test the intrinsic reflectance image generated by the proposed method. For three datasets, given that only one LR-HSI and one PAN image that cover the same area can be used, there are two practicable ways to process HSIs tasks: one is to use the single original LR-HSI, and the other is to perform image fusion first to get a fused HR-HSI. In this article, the fusion methods used for comparison are GSA and CNMF, and the reference made a conclusion that GSA can well preserve the spatial information and reduce spectral distortion, and CNMF can get good results in terms of spatial and spectral aspects [21]. In order to make the comparison process fairer, the reflectance images are recovered from all the aforementioned results. Since it is well known that HIID can well perform in producing reflectance images by eliminating the differences within the same material, the HIID step is added after LR-HSI (LR-HSI-HIID), GSA (GSA-HIID), and CNMF results (CNMF-HIID). The parameters for each comparison method are tuned individually according to the authors' suggestions. In addition, the original HR-HSI (ori-HSI) and its reflectance result (ori-HSI-HIID) are included for a comprehensive comparison of Salinas and Botswana datasets.

The number of training samples for all datasets is set to 1%, 5%, and 10% per class. All reported results are the average and standard deviation of ten experiments on randomly selected training and test sets.

B. Experimental Results and Analysis

In this part, the LR-HSI and PAN image pair will be exploited to yield an HR reflectance image by our method, which will be compared with the images obtained by the other six alternative methods. It is clear that, since reflectance only depends on material properties, it is expected to be beneficial for HSIs' classification.

To evaluate the quality of the classification accuracy, two indexes are used in our study.

1) *Overall Accuracy (OA):* It is the percentage of correctly classified samples to the total number of test samples;

2) *Kappa Coefficient of Agreement (Kappa):* It is used to measure the percentage of agreement corrected by the amount of agreement that can be expected by chance.

1) Visual Comparison: Figs. 3–5 show some experimental results as well as input LR-HSI and PAN images. We can see that both GSA and CNMF fusion methods can get high-quality fusion images since there is a small gap between the fusion results and the original HSI in the performance of the spatial and spectral information in Figs. 3 and 4. Moreover, the HIID method is, indeed, helpful to recover the reflectance component from a given HSI as reflectance images decomposed by the HIID method show better consistency within classes than

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TABLE I

DETAILED INFORMATION OF THE THREE CONSIDERED DATASETS

No.	Salinas		Botswana		Yellow River	
	Categories	Samples	Categories	Samples	Categories	Samples
C1	Brocoli_weeds_1	2009	Water	270	Water	7928
C2	Brocoli_weeds_2	3726	Hippo grass	101	Salt marsh	3568
C3	Fallow	1976	Floodplain grasses 1	251	Exposed tidal flat	3400
C4	Fallow_rough_plow	1394	Floodplain grasses 2	215	Tidal flat reeds	916
C5	Fallow_smooth	2678	Reeds 1	269	Suaeda salsa	1148
C6	Stubble	3959	Riparian	269	Tamarix	2164
C7	Celery	3579	Firescar 2	259	Spartina alterniflora	4816
C8	Grapes_untrained	11271	Island Interior	203	Reeds	3224
C9	Soil_vinyard_develop	6203	Acacia woodlands	314		
C10	Corn_senesced_weeds	3278	Acacia shrublands	248		
C11	Lettuce_romaine_4wk	1068	Acacia grasslands	305		
C12	Lettuce_romaine_5wk	1927	Short mopane	181		
C13	Lettuce_romaine_6wk	916	Mixed mopane	268		
C14	Lettuce_romaine_7wk	1070	Exposed soils	95		
C15	Vinyard_untrained	7268				
C16	Vinyard_trellis	1807				

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR THE SALINAS DATASET

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Ori-HSI	87.70±1.85	86.29±2.08	88.20±0.20	86.86±0.22	89.40±0.22	88.19±0.24
Ori-HSI-HIID	93.58±2.16	92.86±2.41	94.13±0.24	93.49±0.27	95.59±0.15	95.09±0.16
LR-HSI	72.99±10.9	69.69±12.6	85.03±2.98	83.35±3.32	87.32±2.08	85.87±2.33
LR-HSI-HIID	84.12±8.54	82.26±9.66	93.37±2.04	92.62±2.27	95.45±1.58	94.94±1.76
GSA	89.43±1.80	88.23±2.00	89.88±0.27	88.73±0.31	91.45±0.12	90.49±0.14
GSA-HIID	93.52±2.31	92.78±2.58	94.09±0.24	93.42±0.26	95.75±0.14	95.26±0.16
CNMF	88.80±1.71	87.52±1.92	88.87±0.20	87.61±0.22	90.12±0.20	88.99±0.22
CNMF-HIID	91.96±2.16	91.05±2.41	92.40±0.12	91.54±0.13	94.05±0.18	93.38±0.21
ESI-IID	**94.70±2.19**	**94.10±2.44**	**95.29±0.18**	**94.76±0.20**	**96.65±0.14**	**96.33±0.15**

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR THE BOTSWANA DATASET

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
Ori-HSI	78.53±7.63	76.71±8.34	78.60±1.56	76.80±1.70	84.57±0.66	83.28±0.72
Ori-HSI-HIID	**88.31±8.39**	**87.32±9.12**	**88.67±1.73**	**87.70±1.89**	93.20±0.95	92.64±1.03
LR-HSI	63.27±18.9	60.24±20.4	65.51±18.1	62.86±19.1	75.87±8.83	73.89±9.51
LR-HSI-HIID	66.14±16.8	63.44±18.1	67.54±16.5	64.94±17.6	80.40±12.4	78.80±13.4
GSA	79.03±11.1	77.26±12.1	81.13±1.43	79.54±1.55	88.03±0.68	87.03±0.74
GSA-HIID	86.97±8.57	85.86±9.32	87.74±1.47	86.71±1.60	92.59±1.46	91.97±1.58
CNMF	78.81±8.82	77.00±9.60	79.99±1.59	78.32±1.73	86.45±0.90	85.32±0.98
CNMF-HIID	84.57±10.5	83.27±11.45	85.09±1.18	83.83±1.28	91.06±1.04	90.32±1.12
ESI-IID	87.44±8.31	86.38±9.02	87.96±2.64	86.94±2.87	**93.28±0.64**	**92.72±0.69**

undecomposed images, whereas the edges between different classes are blurred. It should be noted that one can see from the zoomed area marked in the red box of Figs. 3 and 4 that the reflectance images produced by ESI-IID not only well maintain consistency within classes but also have distinct edges and accurate color.

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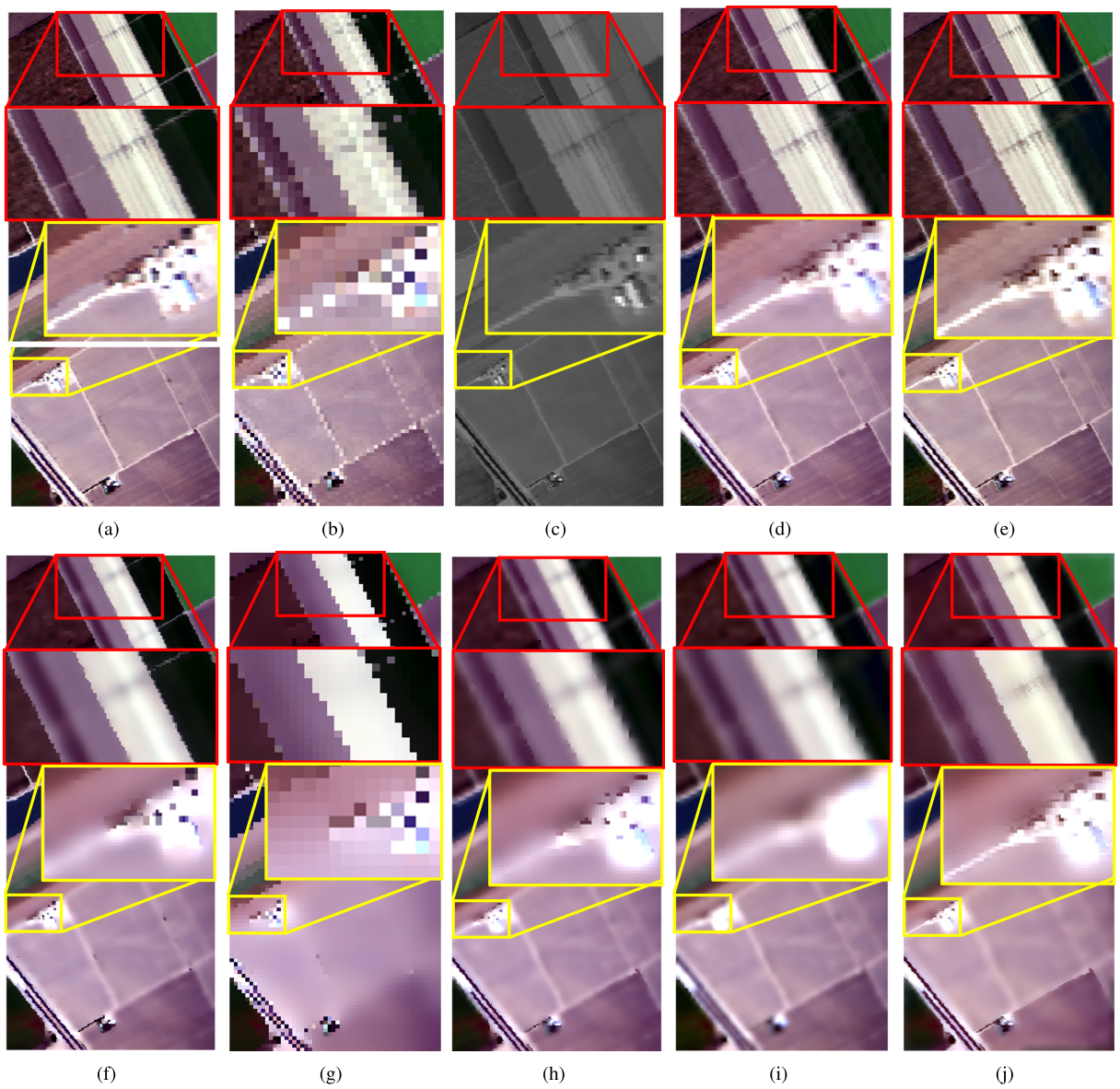


Fig. 3. Fusion and decomposition results comparison for the Salinas dataset. (a) Ori-HSI. (b) LR-HSI. (c) PAN. (d) GSA. (e) CNMF. (f) Ori-HSI-HIID. (g) LR-HSI-HIID. (h) GSA-HIID. (i) CNMF-HIID. (j) ESI-IID.

More specifically, for the Salinas dataset, we can see from the zoomed area marked in the red box of Fig. 3 that both Ori-HSI-HIID and ESI-IID well maintain the intraclass consistency, and the boundary between stubble and fallow classes of Ori-HSI-HIID is clearer than that of ESI-IID. On the contrary, the zoomed area marked in the yellow box shows that ESI-IID performs better than Ori-HSI-HIID on keeping details of the reflectance image. For the Botswana dataset, we can see from the zoomed area marked in the red box of Fig. 4 that the ori-HSI-HIID performs outstandingly in preserving image details obviously; however, it appears to be inferior compared to the ESI-IID method in maintaining the boundary between different classes. In addition, for the Yellow River dataset, we can see from Fig. 5 that there are rich texture structures in the undecomposed HSIs, which is adverse to classification tasks. Moreover, compared to fusion-HIID methods, the ESI-IID provides a better reflectance component containing the least texture information.

It is obvious that the strengths of the ESI-IID method are twofold: 1) it has the same spatial information enhancement capability as the two excellent fusion methods do and 2) different from the IID method for a single HSI, ESI-IID succeeds to produce reflectance images that avoid being oversmoothed of some areas with different materials. In summary, both spatial information enhancement and reflectance recovery problems are considered in the ESI-IID framework, and it succeeds to recover HR reflectance with high quality.

2) Classification Performances Comparison: As reflectance is dependent only on the material properties, it is expected to benefit other HSIs processing tasks, such as HSIs classification. Figs. 6–8 show the classification results of three data sets. As we can see from the red box in Fig. 8(a), the tidal flat

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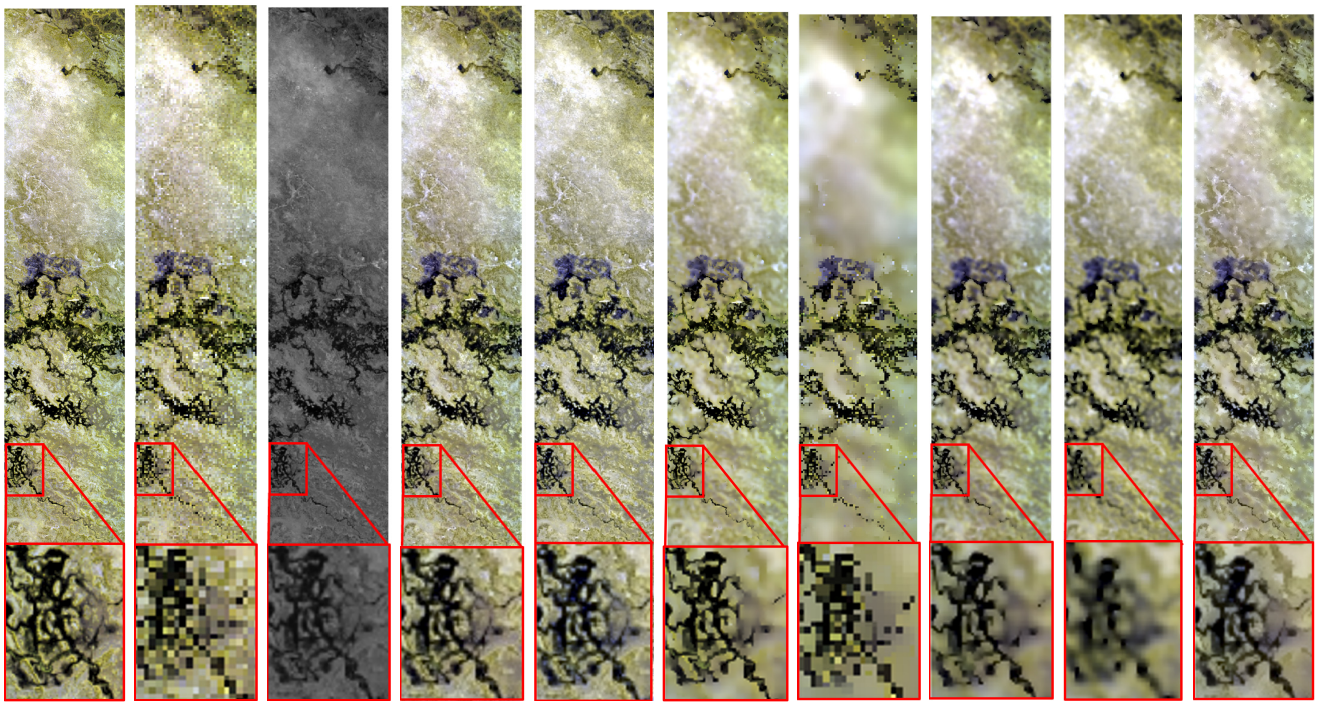


Fig. 4. Fusion and decomposition results comparison for the Botswana dataset. (a) Ori-HSI. (b) LR-HSI. (c) PAN. (d) GSA. (e) CNMF. (f) Ori-HSI-HIID. (g) LR-HSI-HIID. (h) GSA-HIID. (i) CNMF-HIID. (j) ESI-IID.

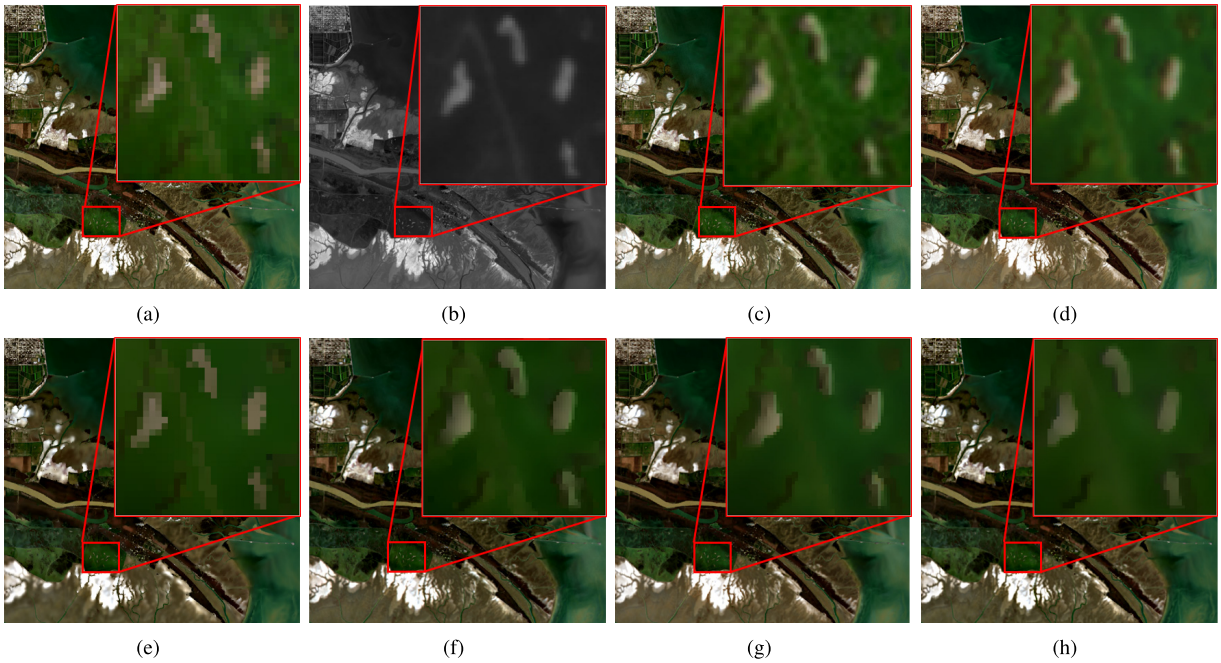


Fig. 5. Fusion and decomposition results comparison for the Yellow River dataset. (a) LR-HSI. (b) PAN. (C) GSA. (d) CNMF. (e) LR-HSI-HIID. (f) GSA-HIID. (g) CNMF-HIID. (h) ESI-IID.

reeds are distributed in a long strip in the middle and a small patch in the lower left corner. Neither the fusion images nor the reflectances recovered from them can perfectly classify the tidal flat reeds, but ESI-IID made it.

On the one hand, regardless of the original HR HSI and its reflectance, one can see from Tables II–IV that the classification results of ESI-IID outperform the other methods for all the datasets, including the simulation datasets and the real dataset. Specifically, compared with the LR-HSIs, our method has an average of approximately 20% and 7% boost for the simulated datasets and the real dataset, respectively. Compared with fusion methods, our method has an average of approximately 5% boost for OA and Kappa. Compared with fusion-HIID methods, our method slightly outperforms

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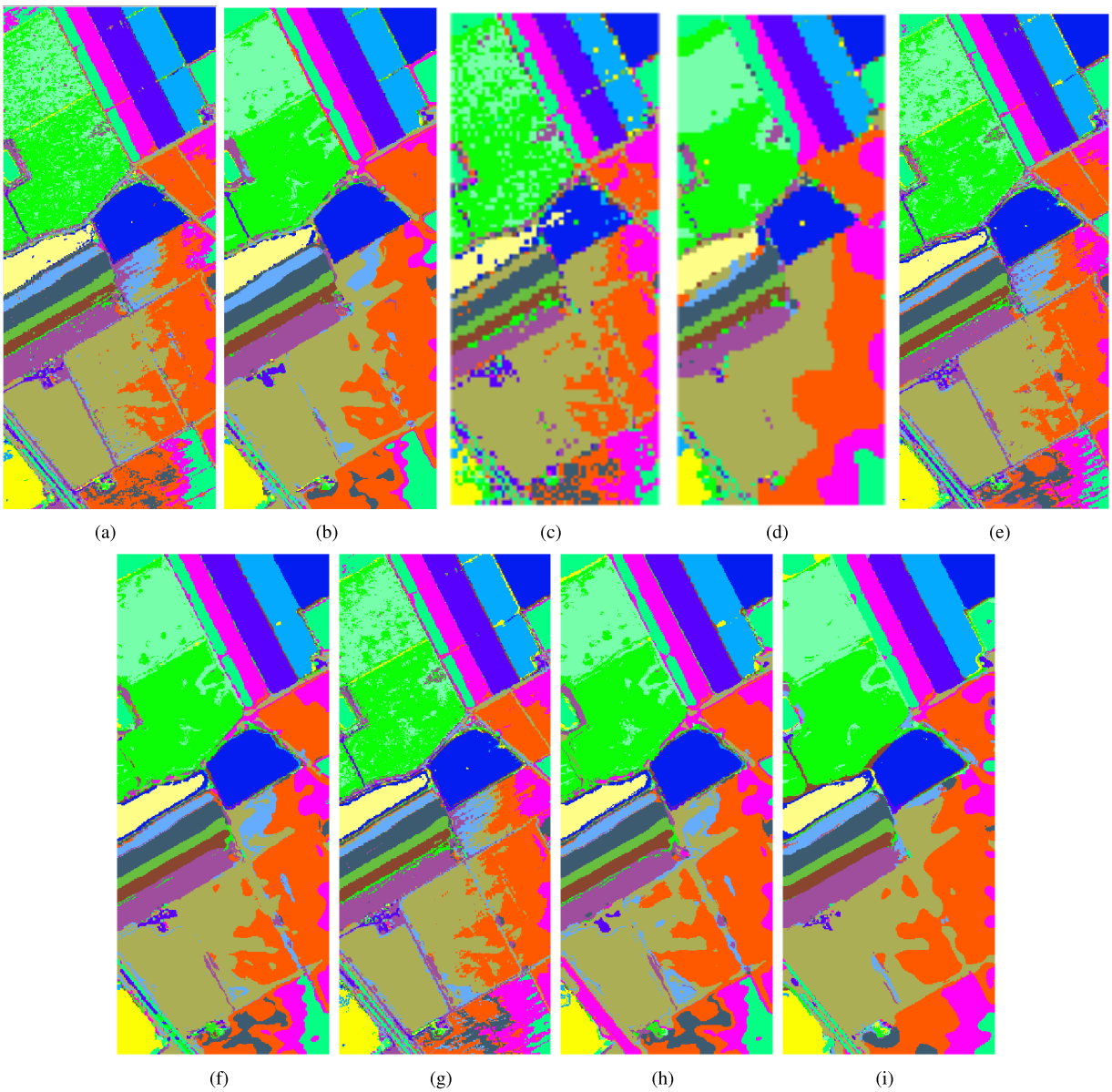


Fig. 6. Classification results of the Salinas dataset. (a) Ori-HSI. (b) Ori-HSI-HIID. (c) LR-HSI. (d) LR-HSI-HIID. (e) GSA. (f) GSA-HIID. (g) CNMF. (h) CNMF-HIID. (i) ESI-IID.

these methods for three datasets. On the other hand, considering the result of ori-HSI and its reflectance, we can see from Table II that ESI-IID slightly outperforms Ori-HSI-HIID for the Salinas dataset in classification performance. For the Botswana dataset, as shown in Table III, the ori-HSI-HIID slightly outperforms ESI-IID when 1% and 5% of the total samples are used as the training samples, while using 10% of samples to train leads to the opposite conclusion. Since the original HR HSI is usually assumed absent, the ori-HSI-HIID and ESI-IID methods are suitable for different application scenes because of each input data. In fact, the experimental results reveal that ESI-IID is competitive with the ori-HSI-HIID method visually and quantitatively. Furthermore, reflectance images can relatively improve the classification accuracy compared with undecomposed images, given that a fake HR-HSI is estimated first. ESI-IID method, however, estimates that the reflectance directly from the original input images has more reliable and better classification performance.

3) *Influence of Threshold τ:* Here, two quantitative indexes are used. One is OA, which is mentioned in Section IV; the other is within-class reflectance variance (WCRV), which can be defined as

$$\mathrm{WCRV}(\mathbf{R}) = \sum_{i=1}^{k} \sum_{j \in C_i} \left\| \mathbf{R}_j - \mathbf{m}_i \right\|_2^2 \tag{29}$$

where k is the total number of categories and $\mathbf{m}_i$ is the mean vector of category C_i in reflectance.

As shown in Fig. 9, we can see that the threshold τ dose has an impact on OA and WCRV. As τ increases, the classification accuracy first improves, then tends to be stable, and, finally,

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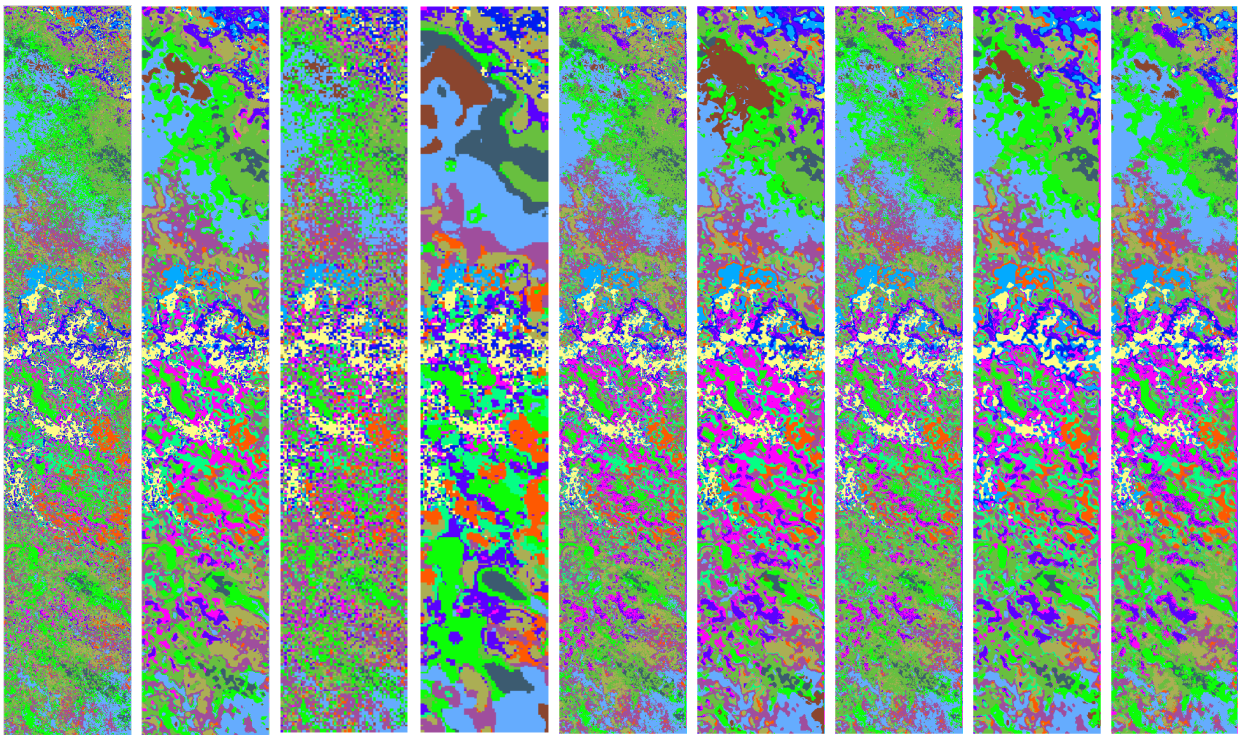


Fig. 7. Classification results of the Botswana dataset. (a) Ori-HSI. (b) Ori-HSI-HIID. (c) LR-HSI. (d) LR-HSI-HIID. (e) GSA. (f) GSA-HIID. (g) CNMF. (h) CNMF-HIID. (i) ESI-IID.

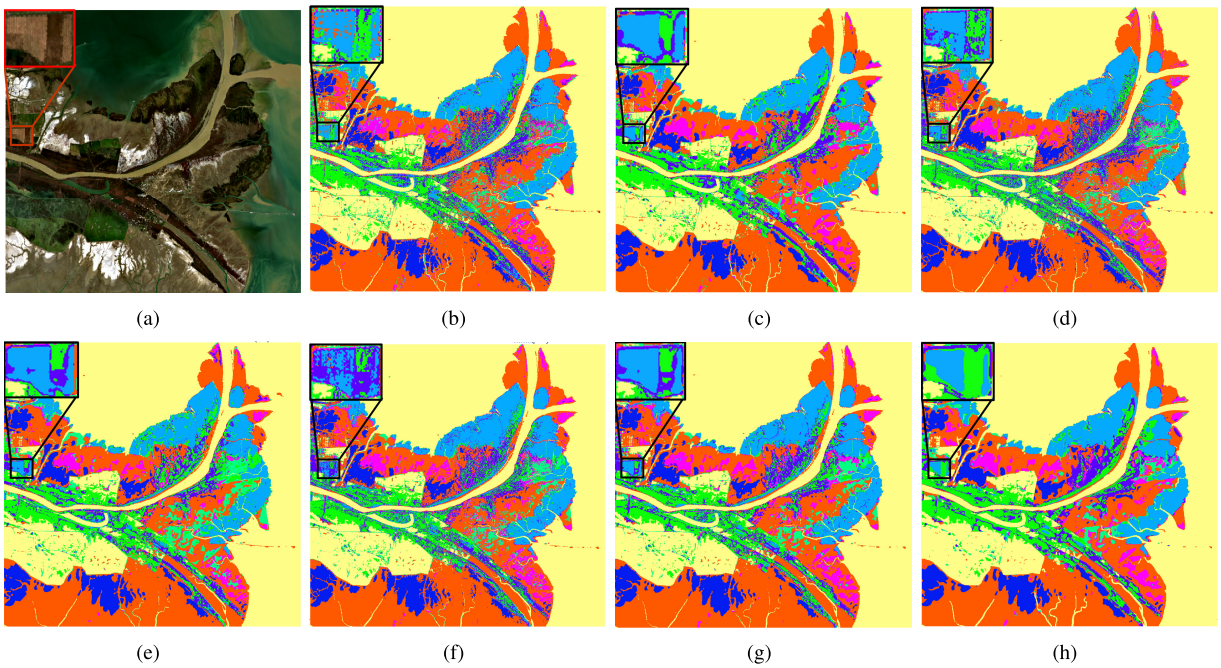


Fig. 8. Classification results of the Yellow River dataset. (a) original LR-HSI. (b) LR-HSI. (c) LR-HSI-HIID. (d) GSA. (e) GSA-HIID. (f) CNMF. (g) CNMF-HIID. (h) ESI-IID.

drops slightly, of which $\tau = 0.04$ is the turning point of the former change. On the contrary, as τ increases, the value of WCRV drops first and then increases, and when τ is equal to 0.02, WCRV becomes relatively optimal. In summary, the value of τ cannot be too large or too small; otherwise, it will lead to poor performance. Therefore, we empirically set τ to 0.06, which means that the classification task performs well, and the WCRV is relatively low.

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH DIFFERENT TRAINING SAMPLES FOR THE YELLOW RIVER DATASET

Methods	1%		5%		10%	
	OA (%)	Kappa (%)	OA (%)	Kappa (%)	OA (%)	Kappa (%)
LR-HSI	89.08±3.14	86.80±3.82	90.09±0.84	88.03±1.00	92.08±0.48	90.42±0.58
LR-HSI-HIID	90.59±3.34	88.64±4.04	91.70±0.83	89.97±0.99	94.70±051	93.59±0.62
GSA	92.97±1.89	91.51±2.29	93.58±0.17	92.23±0.20	95.15±0.28	94.14±0.34
GSA-HIID	95.34±1.63	94.37±1.97	95.87±0.25	95.01±0.30	97.41±0.23	96.87±0.28
CNMF	92.82±1.68	91.33±2.03	92.89±0.36	91.41±0.43	94.26±0.16	93.06±0.19
CNMF-HIID	94.43±2.32	93.27±2.79	95.19±0.27	94.19±0.33	96.60±0.15	95.89±0.18
ESI-IID	**97.16±2.34**	**96.56±2.83**	**97.97±0.42**	**97.55±0.51**	**99.10±0.14**	**98.92±0.17**

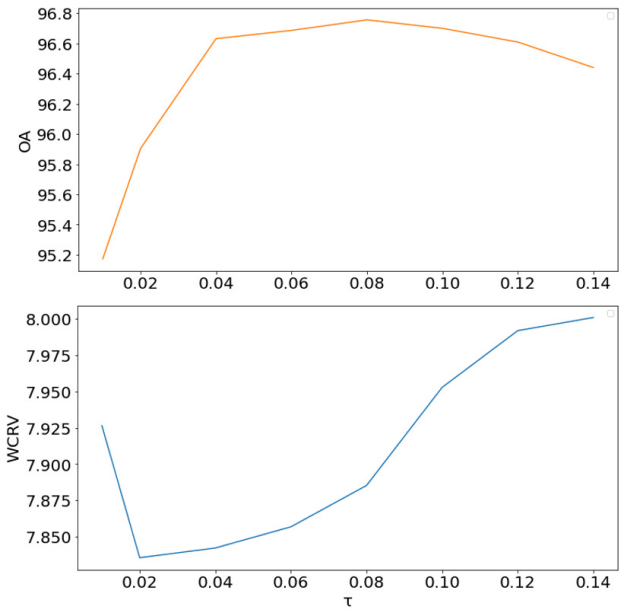


Fig. 9. OA and WCRV with increasing τ for the Salinas dataset.

V. CONCLUSION

In this article, a new HyperIID method called ESI-IID is proposed, which can generate an effective HR intrinsic reflectance from LR-HSI and PAN image pair. The model transforms three reasonable constraints into the problem of finding the minimum Frobenius norm of the proposed matrix energy function, which can be solved iteratively. In the constraint on the reflectance component, by estimating the similarity matrix from real LR-HSI and PAN images rather than estimating that from an artificially generated fused image, our method can recover reflectance images more reliably. The visual and quantitative experimental results on three different hyperspectral datasets demonstrate that the proposed method shows excellent performances.

The future work is to focus on autotuning the threshold parameter and improve the calculation efficiency during the estimation process of the similarity matrix.

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